

Changelog: PerfAudit v1.8

Release Date: July 10, 2026

Build Type: Production / Audit-Ready

[FIXES] Burst Detection Accuracy

- **Time-Based Burst Window:** Replaced the fixed 3-consecutive-sample burst filter with a real-elapsed-time accumulator (using the existing Stopwatch reading). A burst is now flagged only once 300ms of continuous over-threshold time has genuinely elapsed, rather than assuming each sample takes exactly 100ms.
- **Drift Resilience:** Under CPU load, sampling overhead can push real intervals above 100ms. Burst detection is no longer sensitive to this drift, since it now sums actual elapsed milliseconds instead of counting samples.
- **Reset Behavior Preserved:** Any sample that drops back below the configured threshold still resets the accumulator to zero, matching prior behavior.

[NEW] Audit-Ready CSV Reporting

- **Static Computed Summary Values:** Session totals (burst count and average CPU usage) are now calculated directly in C# at export time and written into the CSV as plain numeric values, replacing the previous live Excel formulas (COUNTIF / AVERAGE).
- **Universal Compatibility:** Exported CSV files now display correct summary figures when opened in any tool — Excel, Google Sheets, a text editor, or a Python/pandas script — since the values no longer depend on formula evaluation by the host application.
- **Synchronized Header and Footer:** The top header row (columns F-I) and the bottom summary rows are now populated from the same computed values at the same point in the export flow, eliminating any possibility of the two disagreeing.
- **Clearer Metric Labeling:** Renamed the burst total label from "TOTAL BURSTS DETECTED" to "TOTAL BURST SAMPLES" in both the header and footer, to accurately reflect that the figure counts over-threshold samples rather than discrete burst events.

[UI/UX] Exclusive Fullscreen Awareness

- **Status Bar Notice:** The status label now reads "RECORDING ({threshold}%) - If hidden, target may be in exclusive fullscreen" as soon as a session starts, so testers see the notice before switching back into the game.
- **Tooltip Guidance:** Added a tooltip on the "Always on Top" checkbox clarifying that DirectX/DXGI exclusive fullscreen applications bypass the window manager's z-order, so the PerfAudit window may not remain visible above them even while monitoring continues normally in the background.

[AUDIT] Thread-Safety Verification

- **Crash-Safeguard Review:** Audited the HasExited handling branch inside the precision timer callback. Confirmed all interactive control updates (status label, threshold input, process dropdown) are correctly marshaled to the UI thread via BeginInvoke, and that SafeExport's file-write logic runs independently of its single UI notification call. No code changes were required.

[MAINTENANCE] Codebase Cleanup

- **Comment Removal:** Removed inline and block comments across all project files (Form1.cs, Form1.Designer.cs, Program.cs, and the Properties folder) for a consistent, comment-free codebase, retaining only the single note documenting the exclusive-fullscreen limitation.

[CORE] Preserved "Golden Logic" Features

- **100ms Precision Sampling:** Maintained the high-frequency timer for capturing transient micro-bursts invisible to standard OS tools.
- **Zero-Drift Hardware Timing:** Continued use of System.Diagnostics.Stopwatch for high-resolution sample interval measurement.
- **Kernel-Level Cache Invalidation:** Retained the Process.Refresh() call to bypass .NET's cached process metrics before each read.
- **User-Defined Precision Thresholds:** Retained the v1.7 dynamic threshold input, decimal parsing, and automatic fallback to the 15.0% audit standard on invalid input.